VF-B01 Tensile Strength Properties

Test Method for Tensile Strength Properties	**Test Method #: VF- B01** **REF #: TM-R -044 for textile fabric, TM-R045 for leather, TM-R 046 for laces, TMR-047 for soling**
Date Issued: November 18, 2022 **Revision #: 6**	**Reason for Revision:** **1. Rectify thickness gauge capable of measuring to 0.01mm** **2. Update Figure 1, 2**

PURPOSE AND SCOPE: To measure the following properties:

1. **Tensile strength at break** –the tensile load required to fracture /break a specified cross-sectional area of material.
2. **Elongation at break** –the percentage increase in length of a material at the breaking tensile load.
3. **Elongation at a specified load**- the percentage increase in length of a material at a specified tensile load.
4. **Modulus at a specified elongation** – the tensile load required to stretch a specified cross-sectional area of material to a specified elongation.
5. **Tensile set following at a specified elongation-**the percentage increase in length of a material after being extended to a specified elongation, followed by complete unloading.

REFERENCES:

- ASTM D2209, D2211, D5035, D638, D412, D2256, D1117
- SATRA TM 151, TM137, TM 29, TM 23, TM 94

EQUIPMENT AND MATERIAL:

1. Tensile testing Machine, capable to capture the graphs
2. Clamping jaws, preferably rubber faced.
3. Steel scale, capable of measuring to 0.5mm
4. Thickness gauge, capable of measuring to 0.01mm
5. Vernier caliper, capable of measuring to 0.01mm
6. Cutting dies for each specified material
7. Fabric, Leather (natural and synthetic), rubber, plastic, foam, thread, lace, buckle, paper/board, etc.

SAMPLE PREPARATION:

Condition the sample for at least 48 hours at the standard laboratory condition at 23.0±2.0 ° C Temperature and 55 ±10 % Relative Humidity.

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A. *Fabric:*

1. The test samples should be taken in both the warp and the fill (weft) direction. Samples should be taken such that no two specimens contain the same warp of fill yarn. Samples should always be taken from different areas of the fabric and at least 10cm from any edge.
2. Cut specimens with their long dimensions parallel to the warp (machine) direction and specimens with their long dimensions parallel to the filling (cross) direction. Label the machine direction as direction "A" or "L" and the filling direction as direction "B" or "W"

   - **Cut Strip: non-woven Fabric:**
     - Die cut sample with dimensions of 2.54 ± 0.10cm wide by 20 ± 1cm long with the long dimension accurately parallel to the direction for which the breaking force is required.

   - **Raveled Strip: Woven Fabric**
     - Die cut each specimen 3.5 ± 0.10cm wide by 20 ± 1cm long. Ravel each specimen to give a test width of 2.54 ± 0.10cm by removing an approximate equal number of yarns from each side.
3. With the test sample lays flat, (no bulges of wrinkle), centrally draw two elongation benchmarks at 10 cm apart. These marks /lines should be as thin as possible and clearly drawn across the entire width of the sample, record to the nearest 0.5mm (**D_1**)
4. Measure and record the width of the test sample to the nearest 0.5mm (**W**)

Note: Ribbons and other narrow fabrics, which are 50 mm or less wide, are tested full width. Measure and record the width to the nearest 0.5 mm (W)

B. *Leather:*

1. Test samples should be evaluated in both the length **(L)** and width **(W)** directions. **For leather**, the length direction is parallel to the backbone. **For non-woven backed synthetic leather,** the length direction corresponds to the machine direction.
2. The cutting die should be dumb bell shaped, with a width of 1. 27 ± 0.10cm in the gage area. Refer to Figure 1.
3. With the test sample lays flat, (no bulges of wrinkle), centrally draw two elongation benchmarks at 5 cm apart. These marks /lines should be as thin as possible and clearly drawn across the entire width of the sample. Record to the nearest 0.5mm **(D_1).**
4. Measure and record the width of the test sample to the nearest 0.5mm **(W)**

D. Soling:

1. For test sheet, prepare test specimens cut so that the lengthwise portion of the specimens is parallel to the grain. For specimen from actual soling, cut parallel to the outsole.

   Note:* Some materials are directional *e.g. "thermo" heel counters, extruded plastics, etc.) and testing should be performed in both the machine and cross directions. For directional materials the length direction is parallel to the machine direction.
2. As much as possible, the test samples must be of a uniform thickness particularly on the area of elongation benchmark - thickness range at 3.0 ± 1.0mm.
3. The cutting die should be dumb bell shaped, with a width of 1.0 ± 0.10cm in the gage area. Refer to Figure 2.

4. With the test sample lays flat, centrally draw two elongation benchmarks at 2 cm apart. These marks /lines should be as thin as possible and clearly drawn across the entire width of the sample. Record to the nearest 0.5mm **(D_1).**
5. Measure and record the width of the test sample to the nearest 0.5mm **(W)**
6. Measure the thickness of the sample at each of the elongation bench marks and halfway between the two benchmarks. Record the thinnest of the measurements to the nearest 0.1mm **(T)**

E. **Lace, Thread:**

1. Prepare samples with at least 25 cm length.
2. For laces, ensure that no 2 test samples should be taken from the same lace.
3. For thread, no two test samples should be taken within 10 meters of each other from the same spool. If possible, no two samples should be taken from the same spool. For bobbin thread, samples should always be taken from separate bobbins.
4. With the test sample lays flat, (no bulges of wrinkle), centrally draw two elongation benchmarks at 10 cm apart. These marks /lines should be as thin as possible and clearly drawn across the entire width of the sample. Record to the nearest 0.5mm (D_1).

Note: For specimens that have been abraded prior to this test, the separation should be adjusted to accommodate the specimen size.

F. **Closure Hardware-Buckles, Hooks, Eyelets, Straps, etc.**

1. Thread the corresponding lace of strap through the buckle, hook, of eyelet.
2. Insert the lace or strap in the top jaw of the tensile testing machine.
3. Insert the buckle, hook, of eyelet in the bottom jaw of the tensile testing machine, such that tension can be applied to the components as would normally occur in the finished shoe.

Note: If the buckle, hook, or eyelet is being extracted from a finished shoe, cut the sample such that enough of the upper material remains for secure clamping in the jaws.

REPETITIONS: Test three (3) samples per material. For materials which are direction-dependent, test three (3) samples per material in each direction. For leather laces test (10) samples

GENERAL TEST PROCEDURES:

The following procedures are general test methods for determining tensile properties. Please refer to section below “Requirements for Specific Materials “to obtain the exact requirements for a given material.

A） Tensile Strength and Elongation at break

1. Zero and calibrate the tensile testing machine.
2. Set the tensile testing machine to the specified load range, crosshead speed and jaw gap.
3. Insert the sample centrally in the top and bottom jaws of the tensile testing machine, such that the sample is vertical. No slack or tension is present, and equal portions of the test material are positioned in the top and bottom clamps.

Note: It is generally preferable to use rubber –faced clamping jaws. The use of serrated-faced clamping jaws should be restricted to those materials which require very high clamping pressure and will not be damaged by the serrated face of the jaws.

4. Operate the tensile testing machine at the specified crosshead speed.
5. Follow the distance between the elongation bench marks by placing a steel scale (with gradations of 0.5mm) at one edge of the test sample. Measure the elongation at the inner edges of the two elongation benchmarks (narrowest side) until the breaking (peak) load is obtained. Record the distance to the nearest 0.5mm (**D_2**).
6. Record the breaking load to the nearest 0.1kgf (**B**).

Notes:

- All samples must break at least 1.5cm above the bottom jaw or 1.5cm below the top jaw, any sample not meeting this criterion is considered a “jaw break” and must be disregarded and re-tested.
- For multi-layer materials, (e.g. coated synthetics and leathers) tensile and elongation is measured at the point when the first layer fails.

B) Modulus at a Specified Elongation.

Perform **STEP1-4** as in **Section A** above

1. When the test sample is elongated to the specified amount (often 300%) record the corresponding tensile load as indicated on the tensile testing machine to the nearest 0.1kgf (**M**). In the absence of extensometer use of Vernier calipers set at the specified elongation level is a good tool for exact measurement.

Note: Do not stop the tensile testing machine to obtain the measurement.

C) Elongation at a Specified Load

1. Zero and calibrate the tensile testing machine.
2. Set the testing machine to the specified load range, crosshead speed and jaw gap.
3. Clamp the test specimen in the upper jaw of the tensile testing machine.
   Note: The initial distance between elongation benchmarks must be measured to the nearest 0.01mm (**D_1**) using a Vernier caliper.
4. Load the test sample to the specified load by suspending a weight from the test sample.
   Note: This can be accomplished by centrally attaching a clip or clamp at the bottom of the test sample and hanging a weight from the clamp using a wire or rope. Make sure that the load of the weight is distributed evenly across the entire width of the sample. (The weight of the clamping device must be included as part of the overall loading of the sample, the clamp must be attached below the bottom elongation benchmark.)
5. Measure the elongation between the inner edges of the two elongation benchmarks to the nearest 0.01mm (D_2), with the aid of extensometer. May use a Vernier caliper in the absence of extensometer.

D) Tensile set following a specified elongation

1. Perform **Step 1-4** as in **Section A** above.
2. Stretch the test sample to the specified elongation (often 200%).
3. After 10 minutes, return the tensile testing machine to its starting point.
4. Remove the test sample from the jaws and allow it to relax for 10 minutes.
5. With the test sample lays flat, measure the distance between the inner edges of the two elongation benchmarks to the nearest 0.5mm **(D_2).**

CALCULATIONS:

A. Materials of specified thickness (leather, synthetic leather, fabric etc.)

Tensile Strength (kg/cm) = B / W

Elongation (%) = (D2 – D1) / D1 x 100

B. Materials of unspecified thickness (rubber, plastic, foam etc.)

Tensile strength (kgf/cm^2) = B / W x T

Elongation (%) = (D2- D1) / D1 x 100

Modulus (kgf/cm^2) = M / W x T

REPORT: The average of the three tensile strength measurements should be reported to the nearest 0.1kgf/cm or kgf/cm^2.The average of the three elongation measurements should be reported to the nearest one per cent. The average of the three-modulus measurement should be reported to the nearest 0.1kgf/cm or kgf/cm^2.

Requirements for Specific Materials.

A. Leather, woven & non –woven backed synthetic leather, and paper / board (ASTM D2209 and D 2211)

1. Testing Parameters

Sample size =See Figure 1

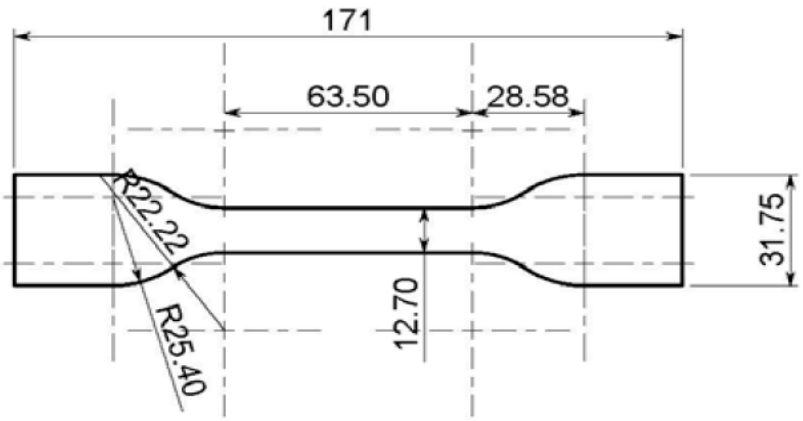
Crosshead speed=250 mm/minute

Jaw gap =10.0 ± 0.3cm

Elongation benchmark gap =5.0 ± 0.1cm

Notes:

a) The (Wet Tensile test) for insole boards is performed by soaking the tensile samples in water at 23 º C for 6 hours Tensile testing is performed immediately upon removing the test samples from the water



Unit: mm

Figure 1

2. Calculation

Tensile strength is reported in kgf/cm width (Calculation A)

B. Fabric (ASTM D5035)

1.Testing parameters

Sample size = 2.54 ± 0.10cm x 20 ± 1cm

Crosshead speed =300 mm/minute

Jaw gap =13.0 ± 0.3cm

Elongation benchmark gap =10.0 ± 0.1cm

2. Calculation

Tensile strength is reported in kgf/cm width. (Calculation A)

C. Plastic, Rubber and Foam (ASTM D638 and D412)

1. Testing Parameters

Sample size =See **Figure 2**

Crosshead speed =500 mm/minute

Jaw gap =6.0 ± 0.3cm

Elongation benchmark gaps =2.0 ± 0.1cm

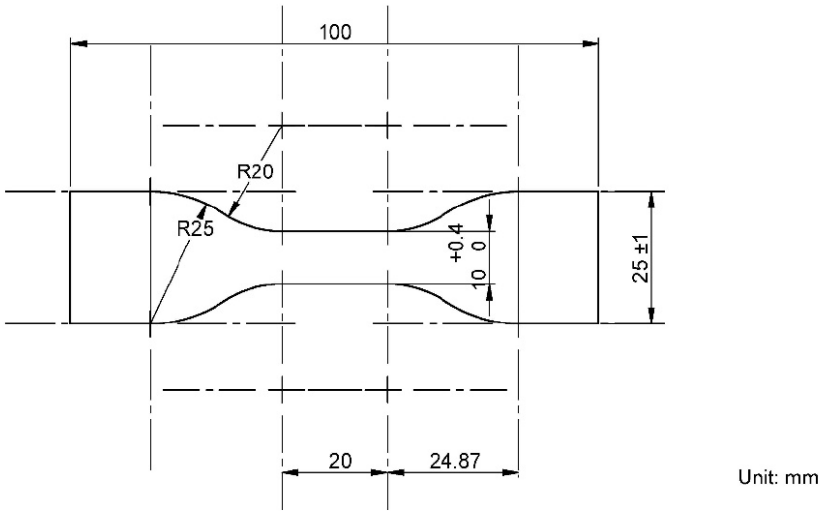


Figure 2

2. Calculation

Tensile strength is generally reported in kgf/cm^2. Calculation B

Note: ***However, for some materials which are always provided in standard thicknesses (e.g. "thermo" heel counters and box toes) the tensile strength is reported in kgf/cm. Calculation A)***

D. Shoelaces (ASTM D5035)

1. Testing Parameters

Sample size = 25 cm minimum

Crosshead speed = 300 mm/minute

Jaw gap =13.0 ± 0.1cm

Elongation benchmark gap =10.0 ± 0.1cm

2. Calculation

Tensile strength is simply reported as the number of kgf required to break the lace.

E. Threads (ASTM D2256)

1. Testing Parameters

Sample size =25 cm minimum

Crosshead speed =300 mm/minute

Jaw Gap =25.0 ± 3.0cm

Elongation benchmark gap =10.0 ± 0.1cm

2. Calculation

Tensile strength is reported in units of kgf, as the cross-sectional area of the thread is not factored into the measurement.

F. Closure Hardware-Buckles, Hooks, Eyelets, Straps, etc. (ASTM D1682)

1. Testing Parameters:

Sample size =25 cm of the maximum amount available

Crosshead speed = 300 mm/minute

Jaw gap =13.0 ± 0.3cm

2. Calculation:

Tensile strength is reported in units of kgf for closure hardware.